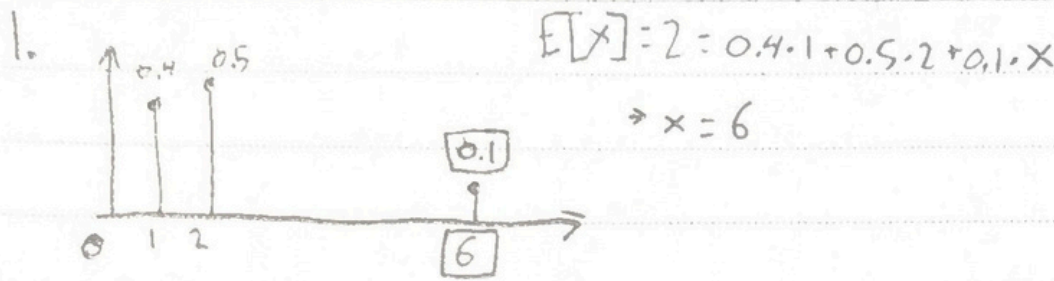


# HW #3



2.  $\text{var}(\alpha) = E[\alpha^2] - (E[\alpha])^2 \Rightarrow 20 = 0.2\alpha^2$   
 $20 = 0.2\alpha^2 - (0.1\alpha^2)$   
 $20 = 0.1\alpha^2$   
 $100 = \alpha^2 \Rightarrow \alpha = 10$

3. a.  $p_x(k) = \left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^{k-1} \quad k=1, 2, 3, \dots$

b.  $E[X] = \frac{1}{\frac{1}{6}} = 6$

4.  $P(0) = \left(\frac{5}{6}\right)^3 = \frac{125}{216}$   
 $P(1) = \binom{3}{1} \left(\frac{1}{6}\right) \left(\frac{5}{6}\right)^2 = \frac{75}{216}$   
 $P(2) = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right) = \frac{15}{216}$   
 $P(3) = \left(\frac{1}{6}\right)^3 = \frac{1}{216}$

5.  $p_Y(x) = \begin{cases} \frac{1}{4}, & Y=4 \\ \frac{1}{2}, & Y=1 \\ \frac{1}{4}, & Y=-2 \end{cases} \Rightarrow E[Y] = \frac{1}{4} \cdot 4 + \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot (-2) = 1$

$\text{var}[Y] = \frac{1}{4}(16) + \frac{1}{2}(1) + \frac{1}{4}(4) - 1 = 4.5$

$E[G] = \frac{125}{216}(-1) + \frac{75}{216}(1) + \frac{15}{216}(2) + \frac{1}{216}(3) = -\frac{17}{216} = -0.079$

6.  $Z = |X - Y| = \begin{cases} \frac{2}{9}, & Z=2 \\ \frac{4}{9}, & Z=1 \\ \frac{3}{9}, & Z=0 \end{cases}$

$E[Z] = \frac{2}{9} \cdot 2 + \frac{4}{9} \cdot 1 = \frac{8}{9}$   
 $\text{var}[Z] = \frac{12}{9} - \frac{64}{81} = \frac{44}{81}$

$E[Z^2] = \frac{2}{9} \cdot 4 + \frac{4}{9} \cdot 1 = \frac{12}{9}$

7. a.  $p_M(k) = \begin{cases} \frac{11}{36}, & k=1 \\ \frac{9}{36}, & k=2 \\ \frac{7}{36}, & k=3 \\ \frac{5}{36}, & k=4 \\ \frac{3}{36}, & k=5 \\ \frac{1}{36}, & k=6 \end{cases}$

8.  $P(C=G \cap D=C) = 0.99^{39} \cdot 0.01 + 0.98^{59} \cdot 0.02 = 0.000041$

b.  $E[M] = \frac{11 \cdot 1 + 9 \cdot 2 + 7 \cdot 3 + 5 \cdot 4 + 3 \cdot 5 + 1 \cdot 6}{36} = 2.53$

$P(C=G \cap D=G) = 0.98^{39} \cdot 0.02 - 0.99^{59} \cdot 0.01 = 0.0000503$

c.  $E[M^2] = 8.36 \quad \text{var}[M] = 8.36 - 2.53^2 = 1.97$

$P(C=G | \text{data}) = \frac{P(\text{data} | C=G)P(C=G)}{P(\text{data})}$   
 $= \frac{\frac{1}{2} \cdot 0.000041}{\frac{1}{2}(0.000041 + 0.0000503)} = 44.94\%$   
 Chris has genuine

d.  $E[M^3] = \frac{11 \cdot 1 + 9 \cdot 2^3 + 7 \cdot 3^3 + 5 \cdot 4^3 + 3 \cdot 5^3 + 1 \cdot 6^3}{36} = 32.86$

e. Geometric,  $E[Y] = \frac{1}{\frac{1}{6}} = 6$

55.06%  
Chris has counterfeit

9. a. geometric, ~~PDF~~  $E[T_n: b] = \frac{1}{\frac{1}{2}} = \boxed{2}$

b.  $E[T_{\text{avg}}] = \sum_{q=1}^{\infty} 2^q \left(\frac{1}{2}\right)^q = \sum_{q=1}^{\infty} 1 = \boxed{\infty \text{ (yiny!)}}$

c.  $E[\text{ord. fid}] = \sum_{q=1}^{11} 2^q \left(\frac{1}{2}\right)^q = \boxed{\$11}$

```

Q = 100000;

% pX(k) will contain the estimated P(X = k)
pX = zeros(1,100);

for q = 1:Q

    x = 0;
    sum = 0;

    % Roll until the running sum reaches 100
    while sum < 100
        roll = ceil(6*rand(1));
        sum = sum + roll;
        x = x + 1;
    end
    % Record the number of rolls
    pX(x) = pX(x) + 1;

end

% Normalize
pX = pX / Q;

stem(1:100,pX);

xlabel('k');
ylabel('p_X(k)');
title('PMF of X');

```

